Module 2 — Synthesis Quiz

Question bank · After exercise lab · open notes · two attempts

Module 2, Synthesis quiz

Readings: Read 0, Read 1, Read 2
Labs: Demo, Exercise
Canvas: Synthesis quiz (Module 2) · **two attempts**, highest score counts

Full integration for Project 2: interpret CIs, read `prop.test()` / `t.test()` output, tidyverse summaries, Type I/II errors in context, and the **five demo-lab CI situations** (mean vs proportion; FPC; small-sample proportion).

Instructions

- Open notes (methods map, R reference).
- Use the CI interpretation template from Read 1: *"I am 95% confident that…"*
- For R items, match the workflow in the demo lab and your Project 2 templates.

Section A — Interpret confidence intervals

Use: *"I am 95% confident that [parameter] in the entire population is between [L] and [U]. In this context, confidence means that if data from many samples were collected the same way, then 95% of the intervals computed from these samples would contain the true parameter."*

A1. First-generation proportion

95% CI for proportion of undergraduates who are first-generation: [**0.22, 0.30**]. Interpret in context.

A2. Website minutes mean

95% CI for mean minutes/day on course websites: [**41, 55**] minutes. Interpret.

A3. Schools older than 50 years

95% CI with FPC for proportion of schools older than 50 years: [**0.14, 0.27**]. Interpret.

A4. Spot incorrect interpretation

Which is the **best** interpretation of a 95% CI?

A) 95% probability the parameter is in this interval.

B) Repeated sampling: ~95% of such intervals contain the parameter.

C) 95% of individuals fall between the endpoints.

Answer key — Section A — Interpret confidence intervals

1. I am 95% confident the true proportion of all undergraduates who are first-generation is between 0.22 and 0.30 (with confidence-level sentence).
2. I am 95% confident the true population mean minutes/day is between 41 and 55 minutes (with confidence-level sentence).
3. I am 95% confident the true proportion of schools older than 50 years in that district is between 0.14 and 0.27 (with confidence-level sentence).
4. **B** is correct. A treats the parameter as random; C confuses parameter with individuals.

Section B — Read R CI output

Run the chunks in the demo lab if needed. Read printed output — do not recompute by hand unless asked.

B1. prop.test output

Context: proportion of CA high school students with a part-time job. `prop.test(x=80, n=220, correct=FALSE)`.

1. Mean or proportion? 2. 95% CI from output? 3. One-sentence interpretation.

B2. t.test output

Context: mean minutes/day on course website. `t.test(x)` on the 25 values from demo lab Problem 1.

1. Mean or proportion? 2. 95% CI? 3. Interpret.

Answer key — Section B — Read R CI output

1. **Proportion.** About [**0.300, 0.428**]. Interpret with template for part-time job proportion.
2. **Mean.** About [**14.6, 22.0**] minutes. Interpret with template for population mean minutes.

Section C — Tidyverse summaries (Read 2 · Project 2 graphs)

These dplyr verbs support mean/median summaries and graphs in Read 2 and Project 2.

C1. Pipe purpose

Main purpose of `x %>% f()` in tidyverse?

C2. filter()

Which function keeps rows satisfying a condition like `hwy > 30`?

C3. select()

Which function chooses columns by name?

C4. summarize()

Which function computes summary statistics (often fewer rows)?

C5. mutate()

Which function adds a new column with the same number of rows?

C6. count()

Which function counts rows per category?

C7. group_by()

Role of `group_by()` before `summarize()`?

C8. n()

Inside `summarize()`, what does `n()` return?

C9. glimpse()

Name two things `glimpse(df)` shows about a dataset.

Answer key — Section C — Tidyverse summaries (Read 2 · Project 2 graphs)

1. **(b)** Chain steps — pass result of one step to the next.
2. **(b)** `filter()`.
3. **(b)** `select()`.
4. **(c)** `summarize()` / `summarise()`.
5. **(c)** `mutate()`.
6. **(a)** `count()`.
7. **(b)** Defines groups so `summarize()` returns one summary **per group**.
8. **(b)** Number of **rows** in the current group.
9. Variable **names**, **types**, dimensions, and preview of values.

Section D — Type I/II errors in context

D1. Lucky coin

Test if coin favors heads (p = P(heads)). State H_0, H_a, Type I error, Type II error in context.

D2. Lucky die

Test if die favors 6. State H_0, H_a, Type I and Type II errors in context.

D3. Medical treatment

Test if treatment effective in **more than 10%** of cases. State hypotheses and both error types.

D4. Left-handed students

Test if school left-handed rate **exceeds 10%**. State hypotheses and both error types.

D5. Defective supplier

Test if defect rate **below 15%**. State hypotheses and both error types.

Answer key — Section D — Type I/II errors in context

1. $H_0 : p = 0.5$, $H_a : p > 0.5$. Type I: claim lucky when fair. Type II: miss biased coin.
2. $H_0 : p = 1/6$, $H_a : p > 1/6$. Type I: claim biased when fair. Type II: miss bias toward 6.
3. $H_0 : p = 0.10$, $H_a : p > 0.10$. Type I: adopt ineffective treatment. Type II: miss effective treatment.
4. $H_0 : p = 0.10$, $H_a : p > 0.10$. Type I: claim higher rate when not. Type II: miss higher rate.
5. $H_0 : p = 0.15$, $H_a : p < 0.15$. Type I: approve high-defect supplier. Type II: reject good supplier.

Section E — Decisions & possible errors

Use $\alpha = 0.05$ for each problem.

E1. 62 heads in 100

$H_0 : p = 0.5$ vs $H_a : p > 0.5$; 62 heads; p-value = 0.0096.

Reject? Type I possible? Type II possible? State Type I in context.

E2. 4 left-handed in 25

$H_0 : p = 0.10$ vs $H_a : p > 0.10$; 4/25 left-handed; p-value = 0.21.

Reject? Type I possible? Type II possible? State Type II in context.

E3. 3 defective in 40

$H_0 : p = 0.15$ vs $H_a : p < 0.15$; 3/40 defective; p-value = 0.034.

Reject? Type I possible? Type II possible? State Type I in context.

Answer key — Section E — Decisions & possible errors

1. **Reject.** Type I **possible** (rejected). Type II **not** (did reject). Type I: claim biased when fair.
2. **Fail to reject.** Type I **not**. Type II **possible**. Type II: miss higher left-handed rate.
3. **Reject.** Type I **possible**. Type II **not**. Type I: claim rate $< 15\%$ when it is not.

Section F — Reducing errors

F1. Reduce Type I

Which helps **reduce Type I error**?

A) Larger α B) Smaller α C) Increase n D) Decrease n

F2. Reduce Type II (fixed α)

With fixed α, what **increases power** / reduces Type II error?

A) Smaller α B) Larger α C) Increase n D) Decrease n

Answer key — Section F — Reducing errors

1. **B** — smaller α requires stronger evidence before rejecting H_0.
2. **C** — larger sample size improves detection when H_0 is false.

Section G — Demo lab · Project 2 CI situations

Match each scenario to the correct interval family and FPC decision. See the demo lab Problems 1–5.

G1. Problem 1 — website minutes

Random sample $n = 25$ minutes/day from a **large** student population.

Parameter? Formula family? FPC?

G2. Problem 2 — classroom chairs

$N = 300$ classrooms, SRS $n = 30$ without replacement; chair counts.

Parameter? FPC?

G3. Problem 3 — part-time job

$n = 220$, $x = 80$ from large population.

Parameter? FPC? Large-count OK?

G4. Problem 4 — schools >50 years

$N = 1500$, SRS $n = 108$, $x = 20$ schools older than 50 years.

Parameter? FPC?

G5. Problem 5 — AP Stats small sample

$n = 10$, $x = 3$ offer AP Stats; rare outcome.

Which method? Why not Wald?

Answer key — Section G — Demo lab · Project 2 CI situations

1. **Mean** μ. t**-interval**, no FPC. `t.test(x)` or $\bar{x} \pm t^* s/\sqrt{n}$.
2. **Mean** μ. t**-interval with FPC** — multiply SE by $\sqrt{(N-n)/(N-1)}$.
3. **Proportion** p. **No FPC**. Large counts met — Wald/`prop.test` OK.
4. **Proportion** p. **Use FPC** on proportion SE; check large counts.
5. **Exact binomial** (`binom.test`). Wald fails: $n\hat{p} < 10$ (and/or $n(1-\hat{p}) < 10$).
